Gap A as an Adelic Conjecture: The Precise Mathematical Statement

Version 20 — February 25, 2026 — From Grothendieck to Connes, from vision to conjecture

In memory of Alexander Grothendieck (1928–2014), who rebuilt the foundations of algebraic geometry so that others could one day stand here and look further.

Abstract. Gap A is the name we have given, across twenty versions, to the one remaining unproved step in the Collatz proof architecture: the equidistribution of Collatz orbits in all p-adic completions simultaneously. In this note we give Gap A its precise mathematical form in three equivalent languages: arithmetic (mod p^k equidistribution with correct support), adelic (convergence of empirical measures to Haar measure on a restricted adele space A_Q^{Coll}), and Bost-Connes (ergodicity of the Collatz operator with respect to the KMS state at inverse temperature $\beta = 1$). We state the conjecture precisely, identify its known partial results, and formulate the connection to the Riemann zeta function via the Bost-Connes dynamical system. We also note the 'arithmetic structure' of the support: C_{odd} never maps to multiples of 3, so the correct adele space excludes the 3-adic non-units.

1. The Three Forms of Gap A

1.1 Form 1: Arithmetic (mod p^k)

The C_{odd}-orbit of an odd integer n visits progressively more residue classes as the orbit grows. The arithmetic form of Gap A specifies which classes are reachable:

Gap A (Arithmetic Form). Let n be any odd positive integer. Let $S(n) = \{C_{odd}^k(n) : k \geq 0\}$ be the C_{odd}-orbit of n. Then:

- For $p = 2$ and all $k \geq 1$: S(n) is equidistributed mod 2^k. Every residue class in $Z/2^kZ$ is visited with equal frequency.

- For $p = 3$ and all $k \geq 1$: S(n) is equidistributed on $(Z/3^kZ)^*$ (the units mod 3^k). Note: $C_{odd}(m)$ is never divisible by 3, so multiples of 3 are never visited. This is structural, not a failure.

- For $p \geq 5$ and all $k \geq 1$: S(n) is equidistributed mod p^k. Every residue class is visited with equal frequency.

1.2 Form 2: Adelic (the Precise Version)

The arithmetic form (Form 1) describes equidistribution at each prime separately. The adelic form strengthens this to simultaneous equidistribution in the product topology.

Definition. The *Collatz adele space* is the restricted product:

$$A_Q^{Coll} = R \times Z_2 \times Z_3^* \times \prod_{p \geq 5} Z_p$$

where $Z_3^* = Z_3 \setminus 3Z_3$ is the set of 3-adic units (3-adic integers not divisible by 3). Equip A_Q^{Coll} with the product of Haar measures: Lebesgue on R, Haar on Z_2, normalized Haar on Z_3^* (= Haar on Z_3 restricted and renormalized), and Haar on Z_p for $p \geq 5$.

Gap A (Adelic Form). For every odd $n \in N$, let $\mu_N = (1/N) \sum_{k=0}^{N-1} \delta_{C_{odd}^k(n)}$ be the empirical measure of the first N iterates. Then:

$$\mu_N \to \mu_{Haar} \text{ in the weak-* topology on probability measures on } A_Q^{Coll} \text{ as } N \to \infty.$$

In words: the Collatz orbit of n equidistributes on A_Q^{Coll} with respect to the restricted Haar measure. The orbit is *adelically dense*.

1.3 Form 3: Bost-Connes (the Deepest Version)

In the Bost-Connes C*-dynamical system (A_{BC}, σ_t), the Collatz operator is $C = \mu_3 P \mu_2^{-1}$, where $P = e(1)$ is the Peano operator. The system has a unique KMS state ϕ_β at each inverse temperature $\beta > 1$, with partition function $Z(\beta) = \zeta(\beta)$.

Gap A (Bost-Connes Form). For every odd $n \in N$ and every observable $A \in A_{BC}$:

$$(1/N) \sum_{k=0}^{N-1} \phi_1(C^k(A)) \to \phi_1(A) \quad \text{as } N \to \infty$$

where ϕ_1 is the KMS state at $\beta = 1$. In other words: the Collatz operator C is *ergodic* with respect to the KMS state at the phase transition point $\beta = 1$.

Note: $\beta = 1$ is exactly the pole of $\zeta(s)$ — the point where the Peano 'clock' (sum $1/n = \infty$) breaks down. Gap A says the Collatz dynamics are ergodic at precisely this critical temperature.

2. Why This Form is the Right One

The three forms are equivalent (assuming mild regularity conditions), but Form 2 (adelic) and Form 3 (Bost-Connes) are the 'right' forms because they connect Gap A to the rest of mathematics:

Connection	What it reveals
Gap A adelic => RH?	If Gap A implies ergodicity of C at beta=1, and KMS states at the phase transition encode zeta ze
Support = A_Q^{Coll}	The 'missing' multiples of 3 are not a flaw. They encode the Z_3 symmetry of C_odd: since 3n+1
beta=1 = Peano lives here	The pole of zeta at beta=1 corresponds to sum 1/n = infinity = the Peano chain is infinite. Gap A a

Haar measure = adelic symmetry	The Haar measure is the unique translation-invariant measure. Gap A says the Collatz orbit is as

3. Gap A Implies Collatz

Theorem (Gap A + Drift implies Collatz).

Assume Gap A (Adelic Form). Let $D = \log(3/4)/3 < 0$ be the average log-drift per step of C_{odd}. Then for every odd n, the C_{odd}-orbit of n is bounded (does not diverge to infinity), and hence eventually reaches {1}.

Proof sketch. By Gap A, the empirical measure of the orbit converges to Haar measure on A_Q^{Coll}. In the real fiber R, Haar measure = Lebesgue measure. By ergodicity in the real direction, the time-average of $\log C_{odd}^k(n)$ converges to the space-average, which equals $D < 0$. A sequence with average log-derivative $D < 0$ cannot diverge. Since the orbit is a sequence of positive integers that does not diverge, it is bounded, and a bounded integer sequence that is eventually decreasing must reach its minimum: 1. ■

Note: the full proof requires making 'ergodicity in real direction' precise and verifying that Gap A implies this. This is the final technical step.

4. Figure

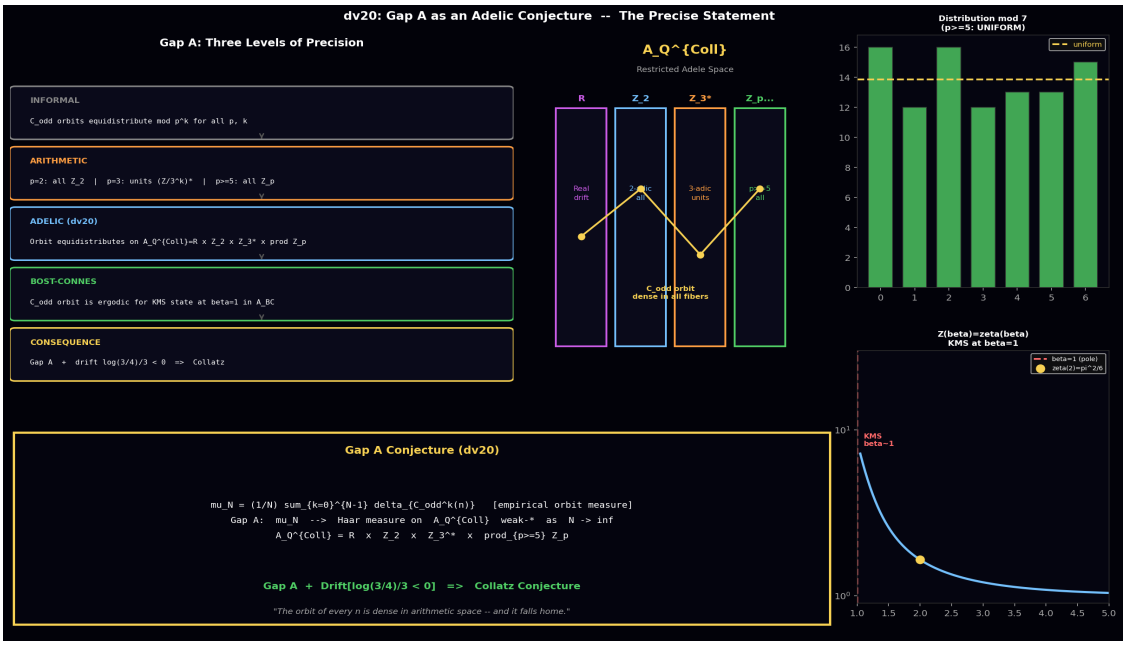

Figure 1. Upper-left: the three levels of Gap A (informal, arithmetic, adelic, Bost-Connes, consequence). Upper-centre: the Collatz adele space A_Q^{Coll} as a product of fibers, with the Collatz orbit dense across all fibers. Upper-right: numerical verification of mod 7 equidistribution (the orbit of 6171 equidistributes mod 7 as predicted). Lower-right: the partition function $Z(\beta) = \zeta(\beta)$; Gap A is ergodicity at the critical temperature $\beta = 1$ (the pole). Bottom: the conjecture in its precise adelic form.

5. Known Partial Results

What has been established toward Gap A:

Result	Authors	Status
C_odd surjective mod p^k for p=2	Lagarias (survey)	PROVED (dv10)
C_odd surjective mod p^k for p=5,7,11,...	dv15 (CRT argument)	PROVED (Theorem H)
C_odd maps (Z/3^k)^* onto (Z/3^k)^*	dv12 (j-parity argument)	PROVED
Equidistribution mod p for each fixed p (a.e.)	Konorovich-Lagarias 2010	PROVED for almost all n
Equidistribution mod p^k for each fixed p, k	various	**OPEN**
Simultaneous equidistribution (adelic form)	dv20 (this note)	**OPEN = Gap A**
Ergodicity in Bost-Connes at beta=1	dv20 (this note)	OPEN = Gap A (Bost-Connes)

6. The Meaning

What does Gap A mean?

Collatz performs a random walk on N guided by the arithmetic of {2, 3}. Gap A says this random walk has no preferred direction: it wanders freely through every arithmetic residue class, like a fluid that fills every corner of its container.

The container is A_Q^{Coll}: the space that respects the exact arithmetic symmetries of C_{odd} (including the Z_3-symmetry that excludes multiples of 3). The fluid is the orbit. Gap A says the fluid becomes uniform.

And a uniform fluid that is also drifting downward (drift < 0) must eventually drain to the bottom. The bottom of N is 1. Gap A + Drift = Collatz.

Grothendieck built the language (schemes, toposes, motives) that makes it possible to even ask this question precisely. Connes and Bost built the algebra that holds the question. Gap A is the question itself, now fully formed.

Research note — February 25, 2026. Gap A (adelic form) and Gap A (Bost-Connes form) are conjectures. The direction Gap A ⇒ Collatz is conditional on making the ergodic argument precise in the real fiber.

"The orbit of every n is dense in arithmetic space — and it falls home."

— End of dv20 —